UNIT 2 ACTIVITY 1:

01_create_database and create_table and insert_record

Code:

```python
import sqlite3

# Connect to the database (creates it if it doesn't exist)
conn = sqlite3.connect("college.db")

# Create a cursor
cursor = conn.cursor()

# Create a table
cursor.execute("""
CREATE TABLE IF NOT EXISTS student(
    id INTEGER PRIMARY KEY,
    name TEXT,
    age INTEGER
)
""")
# Insert records
cursor.execute("INSERT OR IGNORE INTO student VALUES(1,'Tejesh',20)")
cursor.execute("INSERT OR IGNORE INTO student VALUES(2,'Rahul',21)")
cursor.execute("INSERT OR IGNORE INTO student VALUES(3,'Priya',19)")

print("Records Inserted Successfully!")

conn.commit()
conn.close()
```

Output:

```
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr  2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

= RESTART: C:/Users/MY PC/AppData/Local/Programs/Python/Python311/sqlite_demo.py
Table created successfully!

= RESTART: C:/Users/MY PC/AppData/Local/Programs/Python/Python311/sqlite_demo.py
Records Inserted Successfully!

==================================================== RESTART: C:/Users/MY PC/AppData/Local/Programs/Python/Python311/sqlite_demo.py ====================================================
Records Inserted Successfully!
```

02_select_all

```python
print("Records Inserted Successfully!")
# Display all records
print("\nStudent Records")

cursor.execute("SELECT * FROM student")

rows = cursor.fetchall()

for row in rows:
    print(row)

conn.commit()
conn.close()
```

Output:

```
= RESTART: C:/Users/MY PC/AppData/Local/Programs/Python/Python311/sqlite_demo.py
Table created successfully!

= RESTART: C:/Users/MY PC/AppData/Local/Programs/Python/Python311/sqlite_demo.py
Records Inserted Successfully!

================================================== RESTART: C:/Users/MY PC/AppData/Local/Programs/Python/Python311/sqlite_demo.py ==================================================
Records Inserted Successfully!

================================================== RESTART: C:/Users/MY PC/AppData/Local/Programs/Python/Python311/sqlite_demo.py ==================================================
Records Inserted Successfully!

Student Records
(1, 'Tejesh', 20)
(2, 'Rahul', 21)
(3, 'Priya', 19)
```

03_select_one

```python
# Display first record
print("\nFirst Student Record:")

cursor.execute("SELECT * FROM student")

row = cursor.fetchone()

print(row)

conn.commit()
conn.close()
```

Output:

```
================================================== RESTART: C:/Users/MY PC/AppData/Local/Programs/Python/Python311/sqli
Records Inserted Successfully!

Student Records
(1, 'Tejesh', 20)
(2, 'Rahul', 21)
(3, 'Priya', 19)

First Student Record:
(1, 'Tejesh', 20)
```

04_WHERE Clause

```python
# WHERE Clause
print("\nStudent with Age = 21")

cursor.execute("SELECT * FROM student WHERE age=21")

rows = cursor.fetchall()

for row in rows:
    print(row)

conn.commit()
conn.close()
```

Output:

```
Records Inserted Successfully!

Student Records
(1, 'Tejesh', 20)
(2, 'Rahul', 21)
(3, 'Priya', 19)

First Student Record:
(1, 'Tejesh', 20)

Student with Age = 21
(2, 'Rahul', 21)
```

05_ORDER BY

Code:

```
# ORDER BY
print("\nStudents Ordered by Age")

cursor.execute("SELECT * FROM student ORDER BY age ASC")

rows = cursor.fetchall()

for row in rows:
    print(row)

conn.commit()
conn.close()
```

Output:

```
==================================================== RESTART: C:/Users/MY PC/AppData/Local/Programs/Python/Python311/sqlite_demo.py =
Records Inserted Successfully!

Student Records
(1, 'Tejesh', 20)
(2, 'Rahul', 21)
(3, 'Priya', 19)

First Student Record:
(1, 'Tejesh', 20)

Student with Age = 21
(2, 'Rahul', 21)

Students Ordered by Age
(3, 'Priya', 19)
(1, 'Tejesh', 20)
(2, 'Rahul', 21)
```

06_UPDATE Record

Code:

```python
# Update Record
cursor.execute("""
UPDATE student
SET age = 22
WHERE id = 1
""")

print("\nRecord Updated Successfully!")

# Display records after update
print("\nStudent Records After Update:")

cursor.execute("SELECT * FROM student")

rows = cursor.fetchall()

for row in rows:
    print(row)

conn.commit()
conn.close()
```

Output:

```
==================================================== RESTART: C:/Users/MY PC/AppData/Local/Programs/Python/Python311/sqlite_demo.py ====================================================
Records Inserted Successfully!

Student Records
(1, 'Tejesh', 20)
(2, 'Rahul', 21)
(3, 'Priya', 19)

First Student Record:
(1, 'Tejesh', 20)

Student with Age = 21
(2, 'Rahul', 21)

Students Ordered by Age
(3, 'Priya', 19)
(1, 'Tejesh', 20)
(2, 'Rahul', 21)

Record Updated Successfully!

Student Records After Update:
(1, 'Tejesh', 22)
(2, 'Rahul', 21)
(3, 'Priya', 19)
```

07_DELETE Record

code:

```python
# Delete Record
cursor.execute("""
DELETE FROM student
WHERE id = 3
""")

print("\nRecord Deleted Successfully!")

# Display records after delete
print("\nStudent Records After Delete:")

cursor.execute("SELECT * FROM student")

rows = cursor.fetchall()

for row in rows:
    print(row)

conn.commit()
conn.close()
```

Output:

```
==================================================== RESTART: C:/Users/MY PC/AppData/Local/Programs/Python/Python311/sqlite_demo.py ====================================
Records Inserted Successfully!

Student Records
(1, 'Tejesh', 22)
(2, 'Rahul', 21)
(3, 'Priya', 19)

First Student Record:
(1, 'Tejesh', 22)

Student with Age = 21
(2, 'Rahul', 21)

Students Ordered by Age
(3, 'Priya', 19)
(2, 'Rahul', 21)
(1, 'Tejesh', 22)

Record Updated Successfully!

Student Records After Update:
(1, 'Tejesh', 22)
(2, 'Rahul', 21)
(3, 'Priya', 19)

Record Deleted Successfully!

Student Records After Delete:
(1, 'Tejesh', 22)
(2, 'Rahul', 21)
```

08_DROP TABLE

code:

```python
# Drop Table
cursor.execute("DROP TABLE student")

print("\nStudent Table Deleted Successfully!")

conn.commit()
conn.close()
```

Output:

```
Record Deleted Successfully!

Student Records After Delete:
(1, 'Tejesh', 22)
(2, 'Rahul', 21)

Student Table Deleted Successfully!
```

Overall code :

```
import sqlite3


# Connect to the database (creates it if it doesn't exist)

conn = sqlite3.connect("college.db")


# Create a cursor

cursor = conn.cursor()


# Create a table

cursor.execute("""

CREATE TABLE IF NOT EXISTS student(

    id INTEGER PRIMARY KEY,

    name TEXT,

    age INTEGER

)

""")

# Insert records

cursor.execute("INSERT OR IGNORE INTO student VALUES(1,'Tejesh',20)")

cursor.execute("INSERT OR IGNORE INTO student VALUES(2,'Rahul',21)")

cursor.execute("INSERT OR IGNORE INTO student VALUES(3,'Priya',19)")


print("Records Inserted Successfully!")

# Display all records
```

```python
print("\nStudent Records")

cursor.execute("SELECT * FROM student")

rows = cursor.fetchall()

for row in rows:
   print(row)
# Display first record
print("\nFirst Student Record:")

cursor.execute("SELECT * FROM student")

row = cursor.fetchone()

print(row)
# WHERE Clause
print("\nStudent with Age = 21")

cursor.execute("SELECT * FROM student WHERE age=21")

rows = cursor.fetchall()

for row in rows:
   print(row)
# ORDER BY
print("\nStudents Ordered by Age")
```

```
cursor.execute("SELECT * FROM student ORDER BY age ASC")

rows = cursor.fetchall()

for row in rows:
    print(row)
# Update Record
cursor.execute("""
UPDATE student
SET age = 22
WHERE id = 1
""")

print("\nRecord Updated Successfully!")

# Display records after update
print("\nStudent Records After Update:")

cursor.execute("SELECT * FROM student")

rows = cursor.fetchall()

for row in rows:
    print(row)
# Delete Record
cursor.execute("""
```

```
DELETE FROM student

WHERE id = 3

""")


print("\nRecord Deleted Successfully!")


# Display records after delete

print("\nStudent Records After Delete:")


cursor.execute("SELECT * FROM student")


rows = cursor.fetchall()


for row in rows:

   print(row)

# Drop Table

cursor.execute("DROP TABLE student")


print("\nStudent Table Deleted Successfully!")


conn.commit()

conn.close()
```

overall output:

```
Records Inserted Successfully!

Student Records
(1, 'Tejesh', 22)
(2, 'Rahul', 21)
(3, 'Priya', 19)

First Student Record:
(1, 'Tejesh', 22)

Student with Age = 21
(2, 'Rahul', 21)

Students Ordered by Age
(3, 'Priya', 19)
(2, 'Rahul', 21)
(1, 'Tejesh', 22)

Record Updated Successfully!

Student Records After Update:
(1, 'Tejesh', 22)
(2, 'Rahul', 21)
(3, 'Priya', 19)

Record Deleted Successfully!

Student Records After Delete:
(1, 'Tejesh', 22)
(2, 'Rahul', 21)

Student Table Deleted Successfully!
```

Student Skill Tracker application:

Codes:

from flask import Flask, render_template, request, redirect

import sqlite3

```python
app = Flask(__name__)

# Create Database and Table
def create_table():
    conn = sqlite3.connect("student_skill.db")
    cur = conn.cursor()

    cur.execute("""
    CREATE TABLE IF NOT EXISTS students(
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        name TEXT,
        department TEXT,
        skill TEXT,
        level TEXT
    )
    """)

    conn.commit()
    conn.close()

create_table()


@app.route("/")
def home():
```

```
    conn = sqlite3.connect("student_skill.db")
    cur = conn.cursor()

    cur.execute("SELECT * FROM students")
    students = cur.fetchall()

    conn.close()

    return render_template("index.html", students=students)


@app.route("/add", methods=["POST"])
def add():

    name = request.form["name"]
    department = request.form["department"]
    skill = request.form["skill"]
    level = request.form["level"]

    conn = sqlite3.connect("student_skill.db")
    cur = conn.cursor()

    cur.execute(
        "INSERT INTO students(name,department,skill,level) VALUES(?,?,?,?)",
        (name, department, skill, level)
    )
```

```
    conn.commit()
    conn.close()

    return redirect("/")


@app.route("/delete/<int:id>")
def delete(id):

    conn = sqlite3.connect("student_skill.db")
    cur = conn.cursor()

    cur.execute("DELETE FROM students WHERE id=?", (id,))

    conn.commit()
    conn.close()

    return redirect("/")


@app.route("/search", methods=["POST"])
def search():

    skill = request.form["skill"]

    conn = sqlite3.connect("student_skill.db")
    cur = conn.cursor()
```

```
    cur.execute("SELECT * FROM students WHERE skill=?", (skill,))

    students = cur.fetchall()


    conn.close()


    return render_template("index.html", students=students)



if __name__ == "__main__":

    app.run(debug=True)
```

html:

```html
<!DOCTYPE html>

<html>

<head>


<title>Student Skill Tracker</title>


<style>


body{

font-family:Arial;

background:#f2f2f2;

}


.container{
```

```
width:900px;
margin:auto;
background:white;
padding:20px;
margin-top:30px;
border-radius:10px;
box-shadow:0 0 10px gray;
}

h1{
text-align:center;
color:blue;
}

input{
padding:8px;
margin:5px;
width:170px;
}

button{
padding:8px 15px;
background:blue;
color:white;
border:none;
cursor:pointer;
}
```

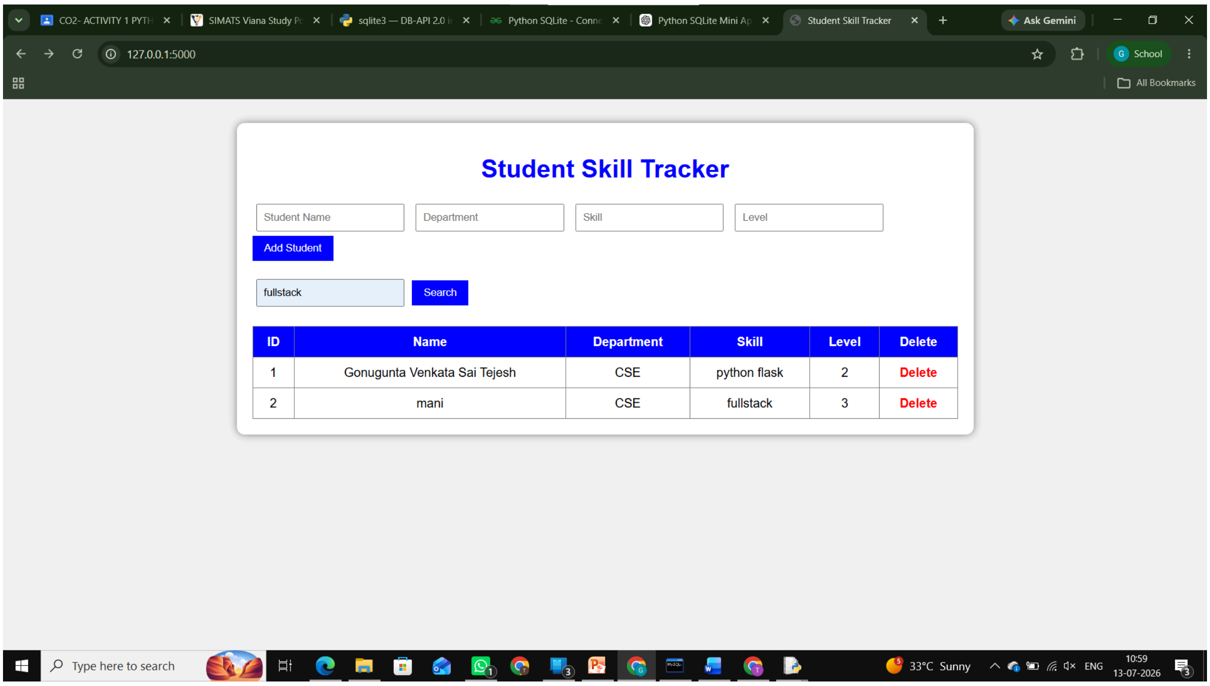
```css
table{
width:100%;
margin-top:20px;
border-collapse:collapse;
}

table,th,td{
border:1px solid gray;
}

th{
background:blue;
color:white;
padding:10px;
}

td{
padding:10px;
text-align:center;
}

a{
color:red;
text-decoration:none;
font-weight:bold;
}
```

```
</style>

</head>

<body>

<div class="container">

<h1>Student Skill Tracker</h1>

<form action="/add" method="post">

<input type="text" name="name" placeholder="Student Name" required>

<input type="text" name="department" placeholder="Department" required>

<input type="text" name="skill" placeholder="Skill" required>

<input type="text" name="level" placeholder="Level" required>

<button>Add Student</button>

</form>

<br>
```

```
<form action="/search" method="post">

<input type="text" name="skill" placeholder="Search Skill">

<button>Search</button>

</form>

<table>

<tr>

<th>ID</th>
<th>Name</th>
<th>Department</th>
<th>Skill</th>
<th>Level</th>
<th>Delete</th>

</tr>

{% for s in students %}

<tr>

<td>{{s[0]}}</td>
```

```
<td>{{s[1]}}</td>

<td>{{s[2]}}</td>

<td>{{s[3]}}</td>

<td>{{s[4]}}</td>

<td>

<a href="/delete/{{s[0]}}">Delete</a>

</td>

</tr>

{% endfor %}

</table>

</div>

</body>
</html>
```

Output:


127.0.0.1:5000
Student Skill Tracker
Student Name
Department
Skill
Level
Add Student
fullstack
Search
ID
Name
Department
Skill
Level
Delete
1
Gonugunta Venkata Sai Tejesh
CSE
python flask
2
Delete
2
mani
CSE
fullstack
3
Delete
Type here to search
33°C Sunny
ENG
10:59
13-07-2026